Methodology Implemented in `causaldosemc`

1 Purpose and Scope

This note specifies the methodology currently implemented in `causaldosemc`. It is a package-specification document rather than a general literature review. The current implementation targets panel data with a continuous treatment that may switch on and off over time, with dose 0 as the unique control state. Active doses may be positive or negative. Unbalanced panels are accepted, but they are internally padded onto the full unit-time grid before estimation.

The package exposes four main methodological layers:

1. `cdmc_fit()`, a matrix-completion baseline estimator with either a staged or joint treatment-effect objective;
2. `cdmc_dr_fit()`, a cross-fitted doubly robust-style linear lag estimator built on out-of-fold baseline prediction and fold-local weighting;
3. `cdmc_dose_response()` and `cdmc_dynamic_estimand()`, which are post-fit functionals of the fitted treatment surface rather than new baseline estimators; and
4. bootstrap, diagnostic, and sensitivity routines that operate on the fitted objects above.

2 Panel Setup and Baseline Structure

Let units be indexed by $i = 1, \ldots, N$ and ordered time periods by $t = 1, \ldots, T$. After internal padding, the package works on a full $N \times T$ grid with observed-sample indicator $O_{it} \in \{0, 1\}$, outcome Y_{it}, dose $D_{it} \in \mathbb{R}$, and optional covariates $X_{it} \in \mathbb{R}^p$. Observed rows must have complete outcome, dose, unit, time, and covariate values.

The untreated outcome surface is modeled as

$$Y_{it}(0) = \alpha_i + \beta_t + X_{it}^\top \gamma + L_{it} + \varepsilon_{it}, \tag{1}$$

where α_i and β_t are additive unit and time effects, γ is the coefficient vector on observed covariates, and $L \in \mathbb{R}^{N \times T}$ is a low-rank latent surface. The package imposes low rank through a nuclear norm penalty.

Given a zero tolerance $\varepsilon_0 \geq 0$ and washout length $w \geq 0$, the eligible-control indicator used by the staged estimator is

$$E_{it} = 1\{O_{it} = 1\}\, 1\{|D_{it}| \leq \varepsilon_0\}\, 1\{\text{no nonzero dose occurs in periods } t-1, \ldots, t-w\}. \tag{2}$$

Thus, a zero-dose observation is excluded from the control sample when it falls within the requested post-treatment washout window.

The implementation requires sufficient optimization support. In particular, the active optimization mask must leave at least one usable cell for every unit and every time period, and the total number of optimization cells must exceed the two-way additive dimension $N + T$.

3 Core Estimator: `cdmc_fit()`

3.1 Staged Objective

The default staged path first estimates the untreated surface on eligible zero-dose observations and then models the residual treatment surface. With optimization weights $w_{it} > 0$ normalized to have mean 1 on the active optimization sample, the stage-1 problem is

$$(\hat{L}, \hat{\alpha}, \hat{\beta}, \hat{\gamma}) \in \arg\min_{L,\alpha,\beta,\gamma} \sum_{i=1}^{N} \sum_{t=1}^{T} E_{it} w_{it} \big(Y_{it} - \alpha_i - \beta_t - X_{it}^{\top}\gamma - L_{it}\big)^2 + \lambda \|L\|_*. \tag{3}$$

The solver alternates between two blocks.

1. **Additive nuisance update.** Conditional on L, the package updates α, β, and γ by alternating weighted row means, weighted column means, and weighted least squares for the covariate block.
2. **Low-rank update.** Conditional on the nuisance fit, the package updates L by singular-value thresholding. Unweighted fits use `softImpute` with SVD thresholding; weighted fits use a proximal weighted singular-value-thresholding step on the same masked squared-error objective.

The outer loop stops when the relative change in the low-rank component falls below the requested tolerance or when the iteration limit is reached.

The fitted baseline counterfactual is

$$\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\beta}_t + X_{it}^{\top}\hat{\gamma} + \hat{L}_{it}. \tag{4}$$

The package then forms the residual treatment surface

$$\hat{\tau}_{it}^{\text{obs}} = Y_{it} - \hat{Y}_{it}(0). \tag{5}$$

For the built-in second stage, let the lag order be $L_d \geq 0$ and define the dose history vector

$$h_{it} = (D_{it}, D_{i,t-1}, \ldots, D_{i,t-L_d})^{\top}. \tag{6}$$

The staged effect layer uses only cells with fully observed lag history and at least one nonzero entry in that history. In the linear specification,

$$g(h_{it}; \theta) = h_{it}^{\top}\theta, \tag{7}$$

and θ is estimated by weighted least squares without an intercept on the sampled residuals $\hat{\tau}_{it}^{\text{obs}}$. In the spline specification, each lag is expanded in a natural-spline basis and recentered so that the fitted effect at the all-zero history is 0.

3.2 Joint Objective

Under `objective = "joint"`, the package augments the optimization problem with the chosen lagged-dose basis and fits baseline and treatment components simultaneously on the full observed estimation sample:

$$(\hat{L}, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\theta}) \in \arg\min_{L,\alpha,\beta,\gamma,\theta} \sum_{i=1}^{N} \sum_{t=1}^{T} O_{it} w_{it} \big(Y_{it} - \alpha_i - \beta_t - X_{it}^{\top}\gamma - B(h_{it})^{\top}\theta - L_{it}\big)^2 + \lambda \|L\|_*. \tag{8}$$

Here $B(h_{it})$ denotes either the raw lag vector or the centered spline basis. For this joint path, pre-sample lag entries are filled with 0 when constructing the treatment basis. Internally, the package solves Equation (8) by appending the treatment-basis columns to the nuisance design and reusing the same alternating low-rank solver.

The returned object separates the fitted baseline counterfactual $\hat{Y}_{it}(0)$ from the fitted treatment component $\hat{\tau}_{it}^{\text{model}} = B(h_{it})^\top \hat{\theta}$.

3.3 Penalty Selection

If the user supplies λ, that value is used directly. Otherwise the package offers two automatic rules.

1. **Heuristic rule.** Compute a residual reference threshold λ_0 after partialling out the additive nuisance block, then set $\lambda = c\lambda_0$ with user-chosen fraction $c \in (0, 1]$.

2. **Blocked cross-validation.** Repeatedly hold out contiguous within-unit blocks from the active optimization sample, fit the stage-1 baseline on the remaining cells, and choose the candidate λ that minimizes holdout mean squared error. If the requested block size is not feasible, the implementation steps down to a smaller contiguous block.

For empirical use, blocked cross-validation is the preferred automatic rule whenever the eligible-control support is sufficiently rich.

4 Cross-Fitted DR Extension: `cdmc_dr_fit()`

The doubly robust layer partitions units into folds $k = 1, \ldots, K$. For each fold, the package fits a weighted `cdmc_fit()` model on the training units, predicts the untreated surface for the holdout units, and then constructs an augmented pseudo outcome on the holdout sample.

The holdout baseline prediction reuses the time-side low-rank basis learned on the training units while estimating holdout-unit coefficients from each holdout unit's own eligible zero-dose observations. Let $\hat{Y}_{it}^{(-k)}(0)$ be the resulting out-of-fold baseline prediction for a holdout cell and define the corresponding observed residual effect

$$\hat{\tau}_{it}^{\text{obs},(-k)} = Y_{it} - \hat{Y}_{it}^{(-k)}(0). \tag{9}$$

The training fold also supplies a linear lag model $h_{it}^\top \hat{\theta}^{(-k)}$ and fold-local normalized weights $w_{it}^{(-k)}$. The package then forms the out-of-fold pseudo outcome

$$\hat{\tau}_{it}^{\text{DR},(-k)} = h_{it}^\top \hat{\theta}^{(-k)} + w_{it}^{(-k)} \left(\hat{\tau}_{it}^{\text{obs},(-k)} - h_{it}^\top \hat{\theta}^{(-k)} \right). \tag{10}$$

After pooling all holdout folds, the final DR coefficients are obtained from an intercept-free linear regression of the pooled pseudo outcomes on the pooled lag history vectors. The built-in DR second stage is therefore linear in the current and lagged doses.

If external weights are supplied, they are passed through the fold-specific stage-1 and augmentation steps. Otherwise the package fits fold-local internal weighting nuisances. The current internal menu includes Gaussian GPS weights, kernel-smoothed GPS weights, CBPS weights, entropy-balance weights, kernel-balance weights, and an adaptive selector over the balancing methods. Internal weights can be stabilized when a GPS density model is used and, by default, are clipped at the smaller of the fold-local 99th percentile and $\sqrt{n}$ for that fold. The package reports fold-level and overall effective sample-size and clipping summaries.

5 Post-Fit Dose Response and Dynamic Functionals

`cdmc_dose_response()` treats the residual treatment surface from a fitted `cdmc_fit()` object as data and estimates a post-fit response model in dose history. This layer does not change the stage-1 matrix-completion baseline. The currently implemented post-fit models are linear, natural-spline, GAM, tree, and forest specifications. For spline, GAM, tree, and forest fits, the response is recentered so that the fitted effect at the all-zero dose history is 0.

`cdmc_dynamic_estimand()` is an evaluation helper rather than a new estimator. Given a fitted `cdmc_fit()`, `cdmc_dr_fit()`, or `cdmc_dose_response()` object and one or more user-supplied dose histories, it returns fitted responses, contrasts against a reference history, or slopes with respect to the contemporaneous dose.

6 Diagnostics, Inference, and Robustness Layers

The package implements several diagnostics around the core estimator.

1. **Placebo refits.** `cdmc_placebo_test()` removes selected zero-dose periods prior to first exposure from the baseline optimization sample, refits the baseline, and computes pseudo effects on those held-out placebo cells.

2. **Carryover screens.** `cdmc_carryover_test()` summarizes residual effects on zero-dose observations occurring a specified number of periods after the most recent nonzero dose. A separate refit-based carryover helper provides the corresponding variant.

3. **Joint placebo and equivalence summaries.** `cdmc_joint_placebo_test()` aggregates placebo windows, and `cdmc_equivalence_test()` implements TOST-style margin checks for placebo or carryover summaries.

4. **SCIA screen.** The SCIA helper compares a restricted current-dose assignment model against an augmented model that adds lagged observed outcomes or lagged residual treatment effects. The resulting incremental F test is a diagnostic for remaining sequential dependence, not a proof of identification.

For inference, `cdmc_bootstrap()` resamples whole unit trajectories with replacement, refits the relevant estimator or diagnostic, and reports percentile intervals for supported summary statistics. When the source fit used automatic penalty selection, the bootstrap can rerun the tuning step inside each replicate so that penalty uncertainty is propagated. Dynamic estimands additionally support max-t simultaneous bands over a finite vector of requested summaries.

For robustness analysis, `cdmc_sensitivity_scan()` replays fits under alternative washout, zero-tolerance, and weighting choices, while `cdmc_sensitivity_bounds()` computes bounded-bias intervals for the supported linear summaries and an optimization-based perturbation layer for objects that require local refitting.

7 Interpretation

The methodology implemented in `causaldosemc` is therefore a structured workflow built around a low-rank untreated-outcome surface, zero-dose masking, distributed-lag treatment-effect modeling, and optional cross-fitted weighting. The package is intentionally modular: the baseline estimator, DR extension, post-fit dose-response layer, diagnostics, bootstrap, and sensitivity tools are separable

but algebraically aligned through the common baseline object $\hat{Y}_{it}(0)$ and the resulting residual treatment surface.